Journal of Russian Laser Research, Volume 17, Number 4, 1996

A STUDY OF ISOTOPICALLY SELECTIVE PHOTOIONIZATION OF YTTERBIUM ATOMS FOR LASER ISOTOPE SEPARATION

S. K. Borisov, M. A. Kuz'mina, and V. A. Mishin

Institute of General Physics, Russian Academy of Sciences, Vavilov St. 38, Moscow 117942, Russia

Abstract

Atomic-vapor laser isotope separation (AVLIS) is studied experimentally and theoretically for ytterbium vapors. The optimum ionization scheme and the process dynamics are determined. The photoionization scheme uses the transitions

$$6^1S_0 \xrightarrow{555.648\ \text{nm}} 6^3P_1 \xrightarrow{581.067\ \text{nm}} (7/2,3/2)_2 \xrightarrow{582.79\ \text{nm}} (52\,353\ \text{cm}^{-1}).$$

For a numerical study of photoionization dynamics, the mathematical model of the AVLIS process is used, which is based on the density matrix formalism and Maxwell's equations. Selective photoionization of the ^{168}Yb isotope is simulated numerically. The yield and selectivity of the process are determined. It is shown that the length of the photoionization region is limited because the laser radiation is absorbed by atoms of ^{171}Yb and ^{173}Yb isotopes. The advantage of the laser method over the method using an electromagnetic separator is demonstrated.

1. Introduction

A new method of isotope separation based on isotopically selective photoionization of atoms has been developed. It is called atomic-vapor laser isotope separation (AVLIS). Studies in this field have been carried out in practically all industrially developed countries, and the advantages are so great that the method is coming into semi-industrial use [1]. The method is based on the difference in absorption spectra of isotopes of one and the same element. It makes use of two- or three-stage selective excitation via intermediate states of atoms of the isotope being separated, and the subsequent photoionization of these atoms from the upper excited state (see Fig. 1a). The advantages of the method are its simplicity and universality. The method can be applied to the majority of elements of the periodic table. Moreover, it can be realized by using high-power dye lasers pumped by copper-vapor lasers (CVLs). Such laser systems are available nowadays.

To understand the range of problems associated with realization of the AVLIS method in practice, let us determine the main stages of the process. First, it is necessary to change the material being separated to the gaseous state. Second, it must be exposed to laser radiation in order to ionize selectively atoms of the given isotope. Finally, the ions formed must be extracted. The realization of the method is hampered by the fact that laser radiation is expensive. Because of this, the action of laser radiation on substance requires thorough study. This study, in its turn, may be divided into two stages. First, the most efficient scheme of excitation and ionization of atoms must be chosen on the basis of certain criteria. Second, the dynamics of the interaction of laser radiation with a medium must be studied in order to determine conditions providing for the optimal use of both the substance being separated and the laser radiation energy. We performed a complete series of works on the study and optimization of isotopically selective photoionization of the rare-abundance (0.14%) ^{168}Yb isotope.

Translated from a manuscript submitted June 25, 1995.

1071-2836/96/1704-0332$15.00 ©1996 Plenum Publishing Corporation

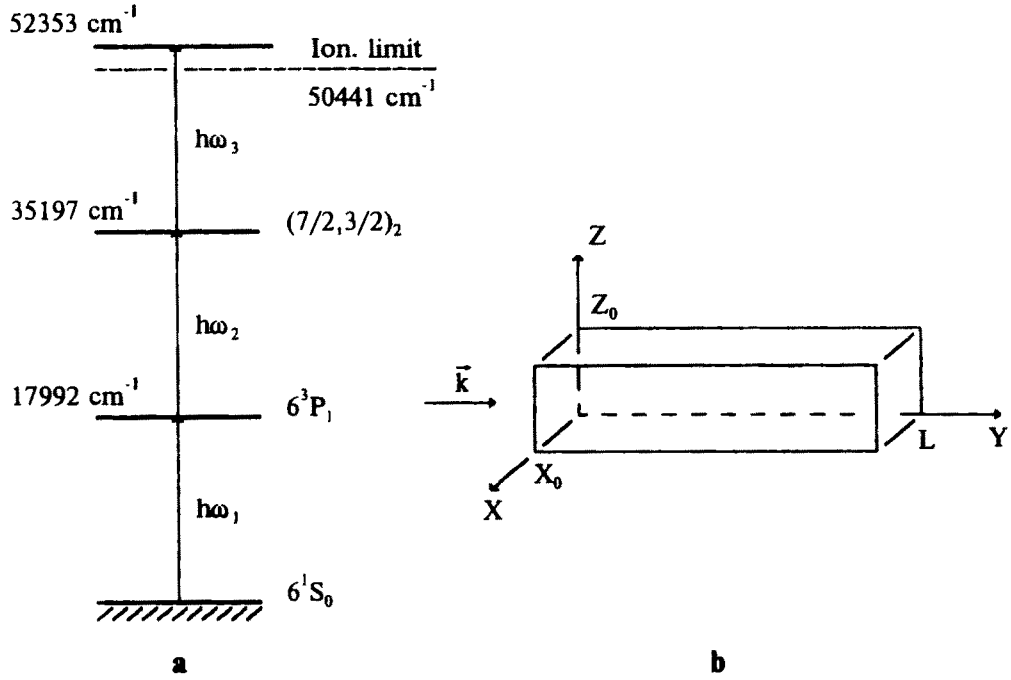


Fig. 1. Optimum scheme of three-stage isotopically selective photoionization of Yb atoms (a); geometry of the problem (b), $\vec{k}$ is the direction of the wave vector of laser radiation.

2. Description of the Scheme Used for Ytterbium Atom Photoionization

The photoionization scheme must satisfy the following criteria: large (in comparison with the Doppler width) isotopic shifts of levels; large lifetimes of intermediate excited states; large cross-sections for autoionization transitions; moreover, it is desirable that operating transitions be found in the wavelength range corresponding to high-efficiency dyes with a long operating life pumped with the green or yellow line of a copper-vapor laser.

We performed this series of studies on the ytterbium atom. Spectroscopic studies included the determination of positions and the identification of energy states, the study of isotopic and hyperfine structures of transitions, the search for autoionization states, the study of radiative characteristics of transitions (lifetimes, dipole moments, cross-sections for autoionization transitions).

The experimental setup designed for carrying out the whole complex of measurements included dye lasers and a vacuum chamber containing the source of the ytterbium atom beam and a quadrupole mass analyzer.

The spectroscopic scheme chosen for selective photoionization of ytterbium atoms used the intercombination transition from the ground $4f^{14}6s^2(6^1S_0)$ state to the $4f^{14}6p6s(6^3P_1)$ state. The 6^3P_1 level has an energy of 17 992.007 cm^{-1} [2]. This transition is excited with radiation at a wavelength of 555.648 nm.

The isotopic structure of the $6^1S_0 - 6^3P_1$ transition of ytterbium was measured with a high accuracy in [3]. The results of identification of the corresponding lines of different isotopes and the isotopic shifts are presented in Table 1. Worthy of mention, in addition to a rather large isotopic splitting of the spectral line of the transition (the total splitting range equals 6225 MHz), is the overlap of lines of the ^{170}Yb and ^{168}Yb isotopes with lines corresponding to the hyperfine structure of transitions for odd isotopes. The latter will have an unfavorable effect on the degree of photoionization selectivity in the course of separation of these

TABLE 1. Isotopic structure of the $6^1S_0 - 6^3P_1$ transition of the ytterbium atom: a.n. is the atomic number of the corresponding isotope; F' and F are the total momenta of the upper and lower levels; $\Delta\nu$ is the isotopic shift of the line of the corresponding isotope relative to the line of the ^{174}Yb isotope

a.n.	$F' - F$	$\Delta\nu$, MHz	a.n.	$F' - F$	$\Delta\nu$, MHz
173	7/2–5/2	–2409	170	1–0	2283
171	1/2–1/2	–2136	173	5/2–5/2	2301
176	1–0	– 948	168	1–0	3636
174	1–0	0	173	3/2–5/2	3804
172	1–0	993	171	3/2–1/2	3816

isotopes.

The lifetime of the 6^3P_1 state of the ytterbium atom was measured in many works (see, e.g., [4]). It is well known and equals 840 ± 20 ns. Knowledge of the lifetime of the 6^3P_1 state makes it possible to determine easily the dipole moment of the $6^1S_0 - 6^3P_1$ transition. Indeed, the lifetime of the 6^3P_1 state unambiguously determines this dipole moment because the transition is isolated. The dipole moment of the $6^1S_0 - 6^3P_1$ was found to be equal to $2.7 \cdot 10^{-30}$ C·m.

For the second stage of the AVLIS process, we chose the transition to the $(7/2,3/2)_2$ level corresponding to the $(4f^{13}6s^26p)$ configuration. The level has an energy of 35 196.98 cm^{-1} [2]. This transition is excited with radiation at a wavelength of 681.067 nm lying in the range of wavelengths at which lasing on the rhodamine 6G dye is obtained.

We were the first to study the isotopic structure of the $6^3P_1 - (7/2,3/2)_2$ transition. The results of identification of the corresponding spectral lines of different isotopes and the isotopic shifts are presented in Table 2. It is worth noting that the spectral line has a large isotopic splitting (its full range equals 13 000 MHz), which is favorable for the degree of selectivity of the separation process. For example, the line of the rare-abundance ^{168}Yb isotope is offset by 1355 MHz from the nearest isotope line. This is promising for obtaining a high coefficient of separation of the isotope mixture for this isotope even under conditions of Doppler broadening of radiative transitions.

Moreover, we measured the lifetime of the $(7/2,3/2)_2$ state of the ytterbium atom. For this purpose, we shut off rapidly the radiation exciting this transition and determined the lifetime directly from the measurements of luminescence dynamics. The $(7/2,3/2)_2$ state of the ytterbium atom was found to have a lifetime of 1.12 ± 0.05 μs. It should be noted that the relatively long lifetime of the $(7/2,3/2)_2$ state of ytterbium is favorable because the spontaneous decay of the $(7/2,3/2)_2$ level competes with the photoionization process (Fig. 1).

The dipole moment of the $6^3P_1 - (7/2,3/2)_2$ transition was measured by the method based on the determination of spectral broadening of the transition line at a high excitation intensity. For a saturation power density of 160 W/cm^2, the line shape was close to the Lorentzian profile with a width of 44 MHz. The dipole moment calculated from these data for the $6^3P_1 - (7/2,3/2)_2$ transition was equal to $4.2 \cdot 10^{-31}$ C·m.

In the following experiments, we looked for autoionization transitions from the $(7/2,3/2)_2$ state of the ytterbium atom and found an intense autoionization resonance at a wavelength of 582.79 nm. The level

TABLE 2. Isotopic structure of the $6^3P_1 - (7/2,3/2)_2$ transition of the ytterbium atom: a.n. is the atomic number of the corresponding isotope; F' and F are the total momenta of the upper and lower levels; $\Delta\nu$ is the isotopic shift of the line of the corresponding isotope relative to the line of the ^{174}Yb isotope

a.n.	$F'-F$	$\Delta\nu$, MHz	a.n.	$F'-F$	$\Delta\nu$, MHz
168	2–1	–10293	173	7/2–5/2	–2532
171	3/2–3/2	– 8805	173	3/2–5/2	–2426
170	2–1	– 6318	173	5/2–5/2	–1833
171	5/2–3/2	– 5973	173	9/2–7/2	– 732
173	1/2–3/2	– 4147	174	2–1	0
173	3/2–3/2	– 3612	173	7/2–7/2	2172
173	5/2–3/2	– 3302	176	2–1	2578
171	3/2–1/2	– 2862	173	5/2–7/2	2894
172	2–1	– 2711			

corresponding to this transition had an energy of 52 353 cm^{-1} and a width of 13 cm^{-1}. It should be noted that the wavelength of radiation exciting this transition corresponds to the maximum of the conversion curve of rhodamine 6G, which is favorable for us. The photoionization cross-section for this autoionization transition was determined from measurements of the saturation power density and was found to be equal to $6.7 \cdot 10^{-16}$ cm^2.

Thus, it may be concluded from the complex study of spectroscopic and radiative characteristics of the ytterbium atom that the scheme of isotopically selective photoionization providing the highest efficiency for laser separation of ytterbium isotopes has the form

$$6^1S_0 \xrightarrow{555.648\ nm} 6^3P_1 \xrightarrow{581.067\ nm} (7/2,3/2)_2 \xrightarrow{582.79\ nm} (52\,353\ cm^{-1}).$$

The diagram of transitions is presented in Fig. 1. In this photoionization scheme, the presence of a strong transition to the autoionization state is well combined with large isotopic shifts for the first and second stages. This photoionization scheme is described in more detail in [5]. The data obtained by us make it possible to study the dynamics of the interaction of laser radiation with the medium in order to optimize the laser separation of ytterbium isotopes.

3. Mathematical Description of the Photoionization Process

To study the dynamics of the interaction of laser radiation with the medium, we concentrated our main efforts on the development of the mathematical model of the AVLIS process. This makes it possible to determine in a short time and inexpensively the ionization efficiency and the physical reasons decreasing the efficiency, and formulate requirements for laser systems, namely, their spectral characteristics and power. As a result, we developed the mathematical model of the AVLIS process. The model correctly fits the conditions

of industrial isotope production and includes both the mathematical description of the process and a package of computer programs.

Let us give a brief description of the model. The model is based on the density matrix formalism and Maxwell's equations. This makes it possible to describe the process most naturally and include into it all parameters of the field and the medium which are of practical importance, namely, inhomogeneous broadening and the degeneration of spectral lines of radiative transitions, a multi-isotopic composition of the medium, quasi-monochromaticity of laser radiation, various relaxation processes, etc. Of considerable importance is the fact that the density matrix formalism makes possible further development of the model because of the absence of intrinsic restrictions.

The evolution dynamics of the medium is described by the Liouville equation

$$i\hbar\left(\frac{d}{dt}+\hat{\Gamma}\right)\hat{\rho}=[\hat{H}_0+\hat{V},\hat{\rho}],$$

where Γ is the operator describing the system relaxation, $\hat{H}_0$ is the Hamiltonian of the unperturbed system, and $\hat{V}$ is the operator describing the interaction of an atomic system with an electromagnetic field. It will suffice to consider the interaction of electromagnetic radiation with an atomic system in a dipole approximation. Because of this, the operator $\hat{V}$ is written in the form $\hat{V}=\hat{d}E/\hbar$, where $\hat{d}$ is the dipole moment operator of the system.

It is assumed in the model that collisions causing longitudinal and transverse relaxations are absent. Therefore, the relaxation of the system is caused only by radiative processes. Let the laser radiation be the superposition of three linearly polarized plane waves, each in resonance with the corresponding transition. Equations for elements of the density matrix were written in the resonance approximation, and Maxwell equations were written in the approximation of slowly varying amplitudes. All these approximations have great validity for all the cases we are interested in. Note also that a rigorous description of the three-stage ionization of atoms requires that the density matrix of dimension 4x4 be used. However, the description can be simplified taking into account that the ionization from the upper excited state may be described incoherently. It may be assumed in this approximation that photoionizing laser radiation introduces only damping into the three-level system, and the rate of damping equals $W_{\text{phi}}=\sigma_3 I_3/\hbar\omega_3$, where σ is the photoionization cross section, I_3 is the intensity of photoionizing radiation, and $\hbar\omega_3$ is the energy of its quantum. This assumption makes it possible to use the density matrix formalism only for the description of the three-level scheme of atom excitation and treat photoionization as one of the relaxation mechanisms described by the usual kinetic balance equation. As for the photoionizing radiation, within the framework of this approach, the Maxwell wave equation may be reduced to the kinetic equation for intensity.

The interaction of quasi-monochromatic laser radiation with an inhomogeneously broadened degenerate system having a multi-isotopic composition is described by the system of equations developed especially for this case (see [6]):

$$\begin{aligned}
\frac{\partial}{\partial t}\rho_{11}^{mi} &= -2\sum_{k=-K}^{K}\left(u_{1k}^{mi}V_{1k}^{mi}+\mu_{1k}^{mi}V_{1k}^{*mi}\right)\\
&\quad +A_{21}^{(m-1)i}\rho_{22}^{(m-1)i}+A_{21}^{mi}\rho_{22}^{mi}+A_{21}^{(m+1)i}\rho_{22}^{(m+1)i};\\
\left(\frac{\partial}{\partial t}+\frac{1}{T_2}\right)\rho_{22}^{mi} &= 2\sum_{k=-K}^{K}\left(\mu_{1k}^{mi}V_{1k}^{*mi}+u_{1k}^{mi}V_{1k}^{mi}-u_{2k}^{mi}V_{2k}^{mi}-\mu_{2k}^{mi}V_{2k}^{*mi}\right)\\
&\quad +A_{32}^{(m-1)i}\rho_{33}^{(m-1)i}+A_{32}^{mi}\rho_{33}^{mi}+A_{32}^{(m+1)i}\rho_{33}^{(m+1)i};\\
\left(\frac{\partial}{\partial t}+\frac{1}{T_3}+W_3\right)\rho_{33}^{mi} &= 2\sum_{k=-K}^{K}\left(\mu_{2k}^{mi}V_{2k}^{*mi}+u_{2k}^{mi}V_{2k}^{mi}\right);
\end{aligned}$$

$$
\begin{aligned}
\left(\frac{\partial}{\partial t}+\frac{1}{2}\frac{1}{T_2}\right) u_{1k}^{mi} &= V_{1k}^{mi}\left(\rho_{11}^{mi}-\rho_{22}^{mi}\right)-V_{2k}^{mi}u_{3k}^{mi}-V_{2k}^{*mi}\mu_{3k}^{mi}\ ;\\
\left(\frac{\partial}{\partial t}+\frac{1}{2}\frac{1}{T_2}\right) \mu_{1k}^{mi} &= V_{1k}^{*mi}\left(\rho_{11}^{mi}-\rho_{22}^{mi}\right)+V_{2k}^{*mi}u_{3k}^{mi}-V_{2k}^{mi}\mu_{3k}^{mi}\ ;\\
\left[\frac{\partial}{\partial t}+\frac{1}{2}\left(\frac{1}{T_2}+\frac{1}{T_3}+W_3\right)\right] u_{2k}^{mi} &= V_{2k}^{mi}\left(\rho_{22}^{mi}-\rho_{33}^{mi}\right)+V_{1k}^{mi}u_{3k}^{mi}+V_{1k}^{*mi}\mu_{3k}^{mi}\ ;\\
\left[\frac{\partial}{\partial t}+\frac{1}{2}\left(\frac{1}{T_2}+\frac{1}{T_3}+W_3\right)\right] \mu_{2k}^{mi} &= V_{2k}^{*mi}\left(\rho_{22}^{mi}-\rho_{33}^{mi}\right)-V_{1k}^{*mi}u_{3k}^{mi}+V_{1k}^{mi}\mu_{3k}^{mi}\ ;\\
\left[\frac{\partial}{\partial t}+\frac{1}{2}\left(\frac{1}{T_3}+W_3\right)\right] u_{3k}^{mi} &= -V_{1k}^{mi}u_{2k}^{mi}+V_{1k}^{*mi}\mu_{2k}^{mi}+V_{2k}^{mi}u_{1k}^{mi}-V_{2k}^{*mi}\mu_{1k}^{mi}\ ;\\
\left[\frac{\partial}{\partial t}+\frac{1}{2}\left(\frac{1}{T_3}+W_3\right)\right] \mu_{3k}^{mi} &= -V_{1k}^{mi}\mu_{2k}^{mi}-V_{1k}^{*mi}u_{2k}^{mi}+V_{2k}^{*mi}u_{1k}^{mi}+V_{2k}^{mi}\mu_{1k}^{mi}\ ;
\end{aligned}
\qquad (1)
$$

$$
\frac{\partial n_+}{\partial t}=W_3 n_3\ ,\qquad W_3=\frac{\sigma_3 I_3(t)}{\hbar\omega_3}\ ,\qquad n_j=\int W(v_y)\rho_{jj}\,dv_y\ ,\quad j=1,2,3\ ;
$$

$$
\begin{aligned}
\frac{\partial \mathcal{E}_{1k}}{\partial y}+\frac{1}{c}\frac{\partial \mathcal{E}_{1k}}{\partial t} &= \sum_i \frac{(\omega_{21}^i)^2 n_0^i}{\varepsilon_0 c\omega_{1k}(2F^i+1)}\int_{-\infty}^{\infty} W(v_y)\sum_{m=-M}^{M} d_1^{mi}\left(\mu_{1k}^{mi}\cos\Delta_{1k}^i+u_{1k}^{mi}\sin\Delta_{1k}^i\right)d(v_y)\ ;\\
\frac{\partial \mathcal{E}_{2k}}{\partial y}+\frac{1}{c}\frac{\partial \mathcal{E}_{2k}}{\partial t} &= \sum_i \frac{(\omega_{32}^i)^2 n_0^i}{\varepsilon_0 c\omega_{1k}(2F^i+1)}\int_{-\infty}^{\infty} W(v_y)\sum_{m=-M}^{M} d_2^{mi}\left(\mu_{2k}^{mi}\cos\Delta_{2k}^i+u_{2k}^{mi}\sin\Delta_{2k}^i\right)d(v_y)\ ;\\
\frac{\partial I_3}{\partial y}+\frac{1}{c}\frac{\partial I_3}{\partial t} &= -\sigma_3 I_3 n_3\ ;
\end{aligned}
$$

$$
V_{1k}^{mi}=\frac{d_1^{mi}\mathcal{E}_{1k}\sin\Delta_{1k}}{2\hbar}\ ;\qquad V_{1k}^{*mi}=\frac{d_1^{mi}\mathcal{E}_{1k}\cos\Delta_{1k}}{2\hbar}\ ;
$$

$$
V_{2k}^{mi}=\frac{d_1^{mi}\mathcal{E}_{1k}\sin\Delta_{2k}}{2\hbar}\ ;\qquad V_{2k}^{*mi}=\frac{d_1^{mi}\mathcal{E}_{1k}\cos\Delta_{2k}}{2\hbar}\ ;
$$

$$
\Delta_{1k}^i=\left[\omega_{21}^i\left(1-\frac{v_y}{c}\right)-\omega_1-k\Delta_{\mathrm{mod}}\right]\left(t-\frac{y}{c}\right)\ ;\qquad \Delta_{2k}^i=\left[\omega_{32}^i\left(1-\frac{v_y}{c}\right)-\omega_2-k\Delta_{\mathrm{mod}}\right]\left(t-\frac{y}{c}\right)\ .
$$

Here, u and μ are the in-phase and quadrature polarization components of the medium, respectively; $n_{1,2,3}$ are the level populations; n_+ is the ion concentration; the Doppler broadening is described by the function $W(v_y)$; $d_{1,2}$ are the dipole moments; ω_{21} and ω_{32} are the transition frequencies; ω_1, ω_2, and ω_3 are the laser frequencies; the index m labels magnetic sublevels; the index i labels atoms of various isotopes; A_{ij} are Einstein coefficients for transitions; T_2 and T_3 are the lifetimes for the second and third levels; F is the total atomic momentum of the ground state; c is the speed of light; ε_0 is the permittivity of vacuum; $\hbar$ is the Planck constant.

Quasi-monochromatic laser radiation is described by a set of equidistant modes with independent phases. Their intensities are given by

$$
I_k(t,\omega_0+k\Delta_{\mathrm{mod}})=I_0(t)\exp\left[-\left(\frac{2k\Delta_{\mathrm{mod}}}{\Delta_{\mathrm{las}}}\right)^2\right]\ ,
$$

where $k=0,\pm1,\pm2,\ldots,\pm K$ (see Fig. 2), Δ_{mod} is the intermode frequency interval, and Δ_{las} is the laser linewidth.

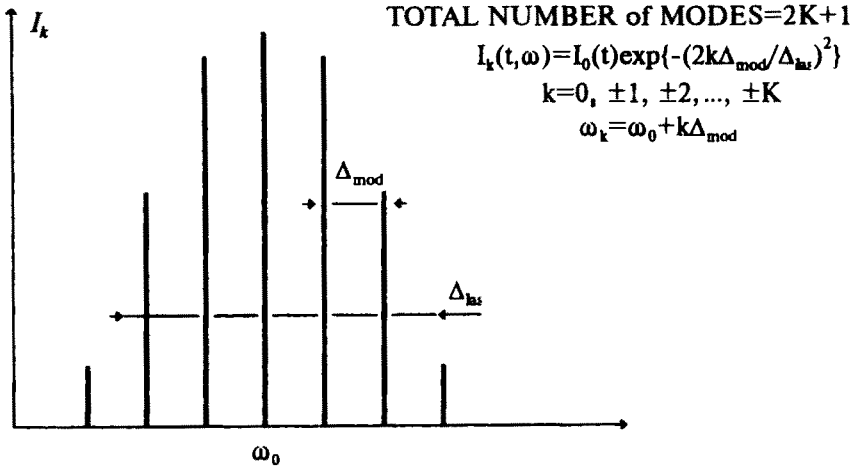


Fig. 2. Schematic diagram of the radiation spectrum of a quasi-monochromatic laser. I_k is the mode intensity; Δ_{las} is the characteristic width of the spectrum; Δ_{mod} is the intermode frequency interval.

A special feature of mathematical simulation of the AVLIS process is that level populations must be calculated for rather large frequency detunings ($10^9 - 10^{10}$ Hz) of the order of the isotopic shift. This is needed for correct calculation of the coefficient of isotope mixture separation and is of particular importance for calculating the separation of isotopes with a low concentration of 0.1–1 %. In principle, Eqs. (1) have a wide domain of applicability and may be used for all those frequency detunings of electromagnetic radiation from radiative transitions which are of interest to us. But attempts to use them for large frequency detunings bring up difficulties associated with features of the numerical solution of these equations. One can see from the form of the equations for the electromagnetic field (EMF) that in this case the solutions represent rapidly oscillating time functions. Because of this, the difference scheme chosen for calculations must have a sufficiently small step, which inevitably increases the number of steps. However, a large number of steps leads to error accumulation and considerably increases the calculation time. Because of this, it is undesirable to use direct numerical integration of equations for large detunings. This difficulty may be obviated by obtaining analytical solutions for the EMF for large detunings. This becomes possible in the case of certain simplifications of (1). First of all it should be noted that for large detunings, the EMF oscillation frequency is dominated by detunings. Because of this, the electric field may be assumed stationary on the scale corresponding to the oscillation frequency. Moreover, we may neglect the contribution of the wings of distant spectral lines of isotopes to the polarization of the medium, assume that they do not affect the evolution of laser pulses, and exclude them from the solution of the self-consistent problem for the medium-field interaction. Because of this, the problem of determining level populations for large detunings is reduced to solving EMF equations without Maxwell's equations, for stationary fields. The same reasoning makes it possible to neglect the contribution of relaxation to the process dynamics. Under these assumptions, the system of equations (1) is easily solved (see [6]). In view of the fact that we are interested in the population of the third level, we present the solution only for ρ_{33}. It has the form

$$\rho_{33} = \frac{V_1^2 V_2^2}{8}\left[\frac{a^2}{C} + \frac{(3a_1^2 - a^2)\cos(|\, a_1 \,|\, t)}{a_1^2(a_1^2 - a_2^2)(a_1^2 - a_3^2)} + \frac{(3a_2^2 - a^2)\cos(|\, a_2 \,|\, t)}{a_2^2(a_2^2 - a_3^2)(a_2^2 - a_1^2)} + \frac{(3a_3^2 - a^2)\cos(|\, a_3 \,|\, t)}{a_3^2(a_3^2 - a_1^2)(a_3^2 - a_2^2)}\right], \quad (2)$$

where a_1^2, a_2^2, and a_3^2 are the roots of the characteristic equation

$$p^6 + 2a^2p^4 + a^4p^2 + C = 0, \quad (3)$$

where

$$
\begin{aligned}
a^2 &= \Omega_1^2 + \Omega_1\Omega_2 + \Omega_2^2 + \frac{3(V_1^2 + V_2^2)}{4} ; \\
C &= \Omega_1^2\Omega_2^2\left[(\Omega_1 + \Omega_2)^2 + \frac{(V_1^2 + V_2^2)}{2}\right] + \Omega_1^4 V_2^2 + \Omega_2^4 V_1^2 \\
&\quad + \frac{\Omega_1^3\Omega_2(4V_2^2 - V_1^2)}{2} + \frac{\Omega_1\Omega_2^3(4V_1^2 - V_2^2)}{2} + \frac{V_1^4(\Omega_1^2/8 - \Omega_1\Omega_2 - \Omega_2^2)}{2} \\
&\quad + \frac{V_2^4(\Omega_2^2/8 - \Omega_1\Omega_2 - \Omega_1^2)}{2} + \frac{V_1^2V_2^2(10\,\Omega_1^2 + 19\,\Omega_1\Omega_2 + 10\,\Omega_2^2)}{8} + \frac{(V_1^2 + V_2^2)^3}{16} .
\end{aligned}
\tag{4}
$$

In the above expression for ρ_{33}, the term making it possible to obtain the time-average values of ρ_{33}

$$
\langle\rho_{33}\rangle = \frac{V_1^2V_2^2a^2}{8C} , \tag{5}
$$

for an arbitrary relationship between the parameters $V_1(t)$, $V_2(t)$, Ω_1, Ω_2 without solving any equations is of particular importance for us. Now, the population of the third level of the ith isotope can be obtained from the expression

$$
n_3^i = n_0^i\frac{V_1^2V_2^2}{8}\int W(v_y)\frac{a^2}{C}\,dv_y , \tag{6}
$$

where

$$
\Omega_1^i = \omega_{21}^i\left(1 - \frac{v_y}{c}\right) - \omega_1 , \qquad \Omega_2^i = \omega_{32}^i\left(1 - \frac{v_y}{c}\right) - \omega_2 .
$$

Here, ω_1 and ω_2 are the laser radiation frequencies and ω_{21}^i, ω_{32}^i are the frequencies of radiative transitions.

Note that the expression obtained for $\langle\rho_{33}\rangle$ may be used not only for the calculation of the coefficient of isotope mixture separation, but for the optimization of the AVLIS process as well.

4. Numerical Experiment

The geometry of the problem is shown in Fig. 1b. It is assumed that an atomic beam travels along the z axis with a thermal velocity. Wave vectors of laser radiation are directed along the y axis. Let us determine the parameters of the problem, namely, typical values of these parameters in actual systems used for laser isotope separation.

Let the density of neutral atoms n in the photoionization region be equal to 10^{13} cm^{-3}. For higher densities, the selectivity of the photoionization process is decreased because of collisional processes. The velocity distribution for v_y components of atoms will be described by the Gaussian function

$$
W(v_y) = \frac{n_0}{\sqrt{\pi}v_0}\exp\left[-\left(\frac{v}{v_0}\right)^2\right] , \qquad v_0 = \sqrt{\frac{2kT}{m_0}} .
$$

In view of the fact that ytterbium is easily evaporated, we took $T = 1000$ K.

The energy density of laser pulses is chosen from the saturation condition for the autoionization transition

$$
E_1 = E_2 = E_3 = \frac{\hbar\omega_3}{\sigma_{\text{phi}}} .
$$

For numerical experiments, we set $E_1 = E_2 = E_3 = 5$ J/m^2.

The laser pulse was assumed to have a Gaussian envelope with half-width $T_0 = 10$ ns:

$$I = \frac{E_j}{\sqrt{\pi} T_0} \exp\left[-\left(\frac{t}{T_0}\right)^2\right] .$$

In our opinion, this pulse shape adequately approximates the actual pulse of a CVL-dye laser system.

Now let us determine some characteristics of the photoionization process determining its selectivity, yield, and efficiency.

The process yield can be described by the expression

$$P^i = m_0^i n_0^i X_0 Z_0 L f_r K_L^i . \tag{7}$$

Here, P^i is the rate of the ith isotope production; m_0^i is the mas of the atom; n_0^i is the density of atoms of the ith isotope in the photoionization region; $X_0 Z_0 L$ is the volume of the photoionization region; f_r is the repetition rate of laser pulses. The degree of ionization of the ith isotope in a medium is characterized by the ionization coefficient K_L^i determined by the expression

$$K_L^i = \frac{1}{L} \int_0^L K^i(y)\, dy , \tag{8}$$

where $K^i(y)$ is the ratio of the concentration of the ith isotope upon the passage of a laser pulse n_+^i to the initial concentration of this isotope n_0^i in the elementary volume $X_0 Z_0\, dy$.

The isotopic selectivity of the separation process is characterized by the separation coefficient α_L defined by the expression

$$\alpha_L^i = \frac{1}{L} \int_0^L \alpha^i(y)\, dy . \tag{9}$$

The quantity α^i is the ratio of the relative concentrations of the isotope being separated prior to and after the separation process in the elementary volume $X_0 Z_0\, dy$:

$$\alpha^i(y) = \frac{n_+^i}{1 - n_+^i} \cdot \left(\frac{n_0^i}{1 - n_0^i}\right)^{-1} .$$

The energy cost of the separation process is determined by the laser radiation energy expended for a unit mass of the isotope separated in the product:

$$\gamma = \frac{E_1 + E_2 + E_3}{M^i} ; \qquad M^i = m_0^i n_0^i L K_L^i . \tag{10}$$

In view of the fact that at least three photons are required to photoionize a single atom, the minimum possible energy cost of the separation process will be defined by the expression

$$\gamma_0 = \frac{\hbar\omega_1 + \hbar\omega_2 + \hbar\omega_3}{m_0} .$$

For the ytterbium photoionization scheme chosen above,

$$\gamma_0 = \frac{1.04 \cdot 10^{-18}}{2.8 \cdot 10^{-22}\ \text{g/J}} = 3.7 \cdot 10^3\ \text{J/g} .$$

It is clear that the energy cost for 1 g of the product under actual conditions will be considerably greater. Thus, the quantity γ_0 represents the natural scale of the energy cost of the process, and the ratio γ_0/γ is the factor characterizing the energy efficiency of the separation process.

The system of equations (1) was used to study the separation of the ^{168}Yb isotope. Laser radiation was assumed to have a continuous power distribution over the spectral line profile at both the first and second stages of the process. It is obvious that the laser radiation linewidths must match the Doppler linewidths for efficient interaction of the laser radiation with the inhomogeneously broadened medium. Because of this, the laser line profile was approximated by a Gaussian function with characteristic width Δ_{las} equal to the Doppler linewidth for the corresponding transition. The spectrum of the laser pulse will be determined by the following model. Let us assume that it consists of separate equidistant radiation modes (they should not be confused with longitudinal laser modes) with uncorrelated phases (see Fig. 2). In the ideal case, we must have an infinite number of modes, which gives a spectrum that is closest to the homogeneous one. However, this is impossible from the viewpoint of numerical calculations. But if the number of modes taken for calculations is finite and the intermode interval is close to $1/T_0$, the effect of radiation with this spectrum on the medium may be thought of as only weakly different from the effect produced by radiation with a continuous power distribution over the line profile. The calculations were performed for 11 modes. The centers of laser lines were tuned exactly to the frequencies of spectral transitions of the ^{168}Yb isotope.

The results of exposure of a medium to laser radiation are illustrated in Figs. 3–7. The dynamics of an inhomogeneously broadened ytterbium medium and velocity distribution functions for populations of the first, second, and third levels averaged over the pulse duration for $y = 0$ are presented in Figs. 3 and 4. One can see that laser radiation interacts with practically all the inhomogeneously broadened system. Owing to the fact that laser radiation is quasi-coherent, absorption is nearly of the classical nature. One can see from Fig. 3 that laser radiation mixes the populations of the first, second, and third levels is such a way that the time-average population of each of these levels is equal to one-third of the total number of particles found in the system. As expected, the ionization efficiency is relatively low in this case and equals 20 %. Figure 5 presents the distributions of the third-level population and the concentration of ions of the ^{168}Yb isotope along the photoionization region upon the passage of laser pulses. One can see that the ionization coefficient continuously decreases with increasing length and virtually vanishes at a length of 4 m. This phenomenon is easily explained. Indeed, at the first stage of the process, the absorption line of the ^{168}Yb isotope overlaps, in view of the Doppler broadening, the spectral lines of the ^{171}Yb and ^{173}Yb isotopes (See Table 1) whose abundance is lower than that of the ^{168}Yb isotope by approximately a factor of 300. Because of this, the absorption of a laser pulse at the first stage is considerably stronger than the absorption of pulses at the second and third stages. This is clearly seen in Figs. 6 and 7. As a result, the photoionization process ceases at the end of the 4-m section of the medium. We see that under the given conditions, the length of the photoionization region is limited by the light-wave absorption by atoms of unseparated isotopes and can be increased only by increasing considerably the radiation power used at the first stage. Moreover, the numerical experiments showed that the ionization coefficient could be increased up to 60 % (within the limits of a 4-m length of the photoionization region) by increasing the photoionizing radiation power up to 20 W.

Now, formula (7) can be used to determine the output and energy efficiency of the process for the ^{168}Yb isotope ($m_0 = 2.8 \cdot 10^{-22}$ g; $n_0 = 1.4 \cdot 10^{10}$ cm^{-3}; $X_0 Z_0 = 1$ cm^2; $L = 400$ cm; $f_r = 10$ kHz; $K_L = 0.18$). We obtain $P^i = 2.8 \cdot 10^{-6}$ g/s or 10 mg/hour. In this case, the total expenditure of average laser radiation power equals 15 W (5 W for every stage of the process). Now from formula (10) it follows that the energy cost of the process is $\gamma = 2.1 \cdot 10^6$ J/g. Comparing this value with the previously determined minimum energy cost of the process $\gamma_0 = 3.7 \cdot 10^3$ J/g, we see that only 1/1800 part of the laser energy is utilized. The main reason for the low laser energy utilization is the absorption of laser radiation at the first stage of the process by atoms of unseparated isotopes. It can be easily shown that the laser energy utilization might be increased by a factor of about 300, provided absorption by these isotopes is absent.

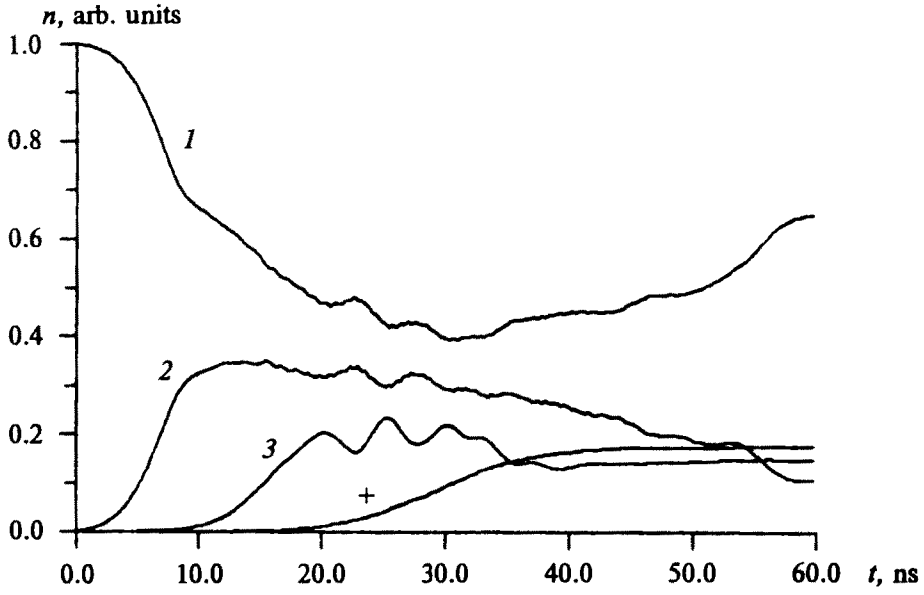


Fig. 3. Dynamics of level populations in the field of quasi-monochromatic laser radiation for $y = 0$: 1, 2, 3) relative concentrations of ^{168}Yb atoms on the first, second, and third levels, +) relative concentration of ions of ^{168}Yb.

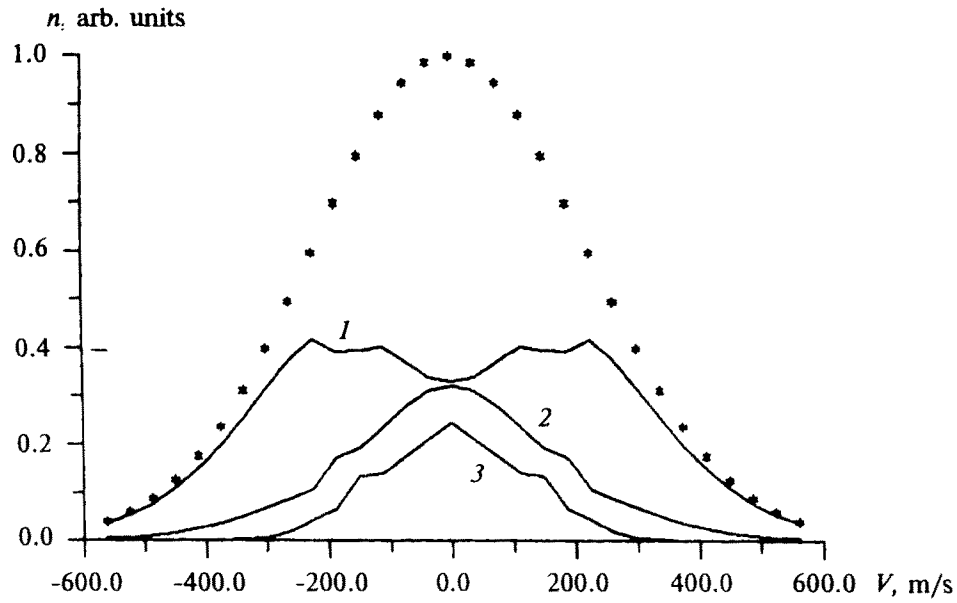


Fig. 4. Velocity distributions of atoms found on the first, second, and third levels prior to and upon the passage of laser radiation.

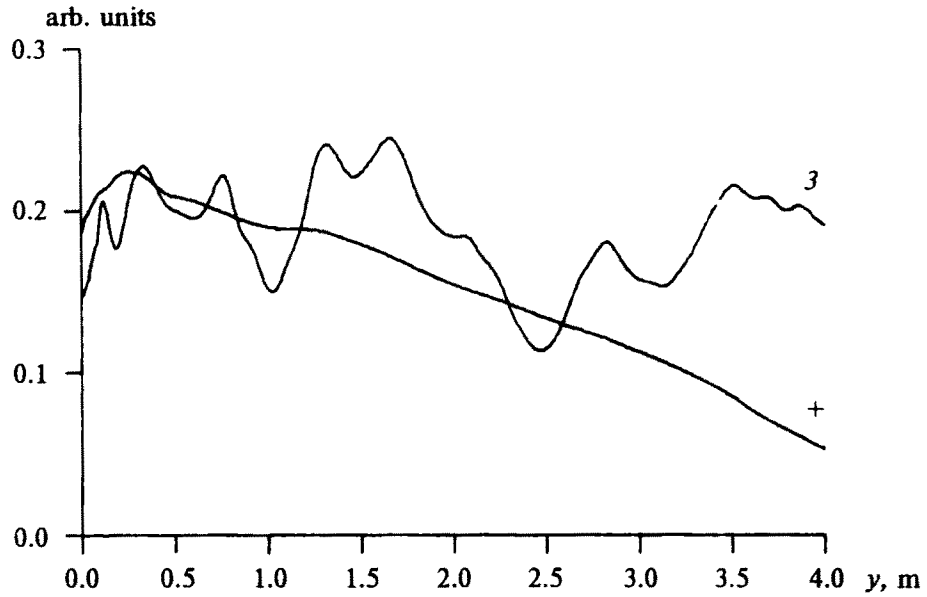


Fig. 5. Distributions of the third-level population and the ion concentration along the photoionization region upon the passage of laser radiation.

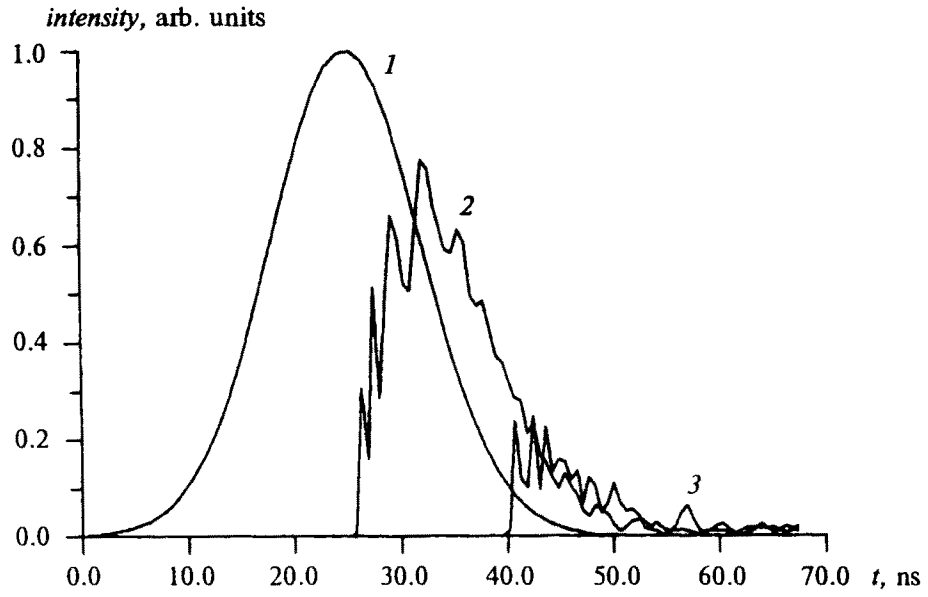


Fig. 6. Shapes of the laser pulses used at the first stage of the process at three points of the medium $y = 0$, $y = 2$ m, and $y = 4$ m.

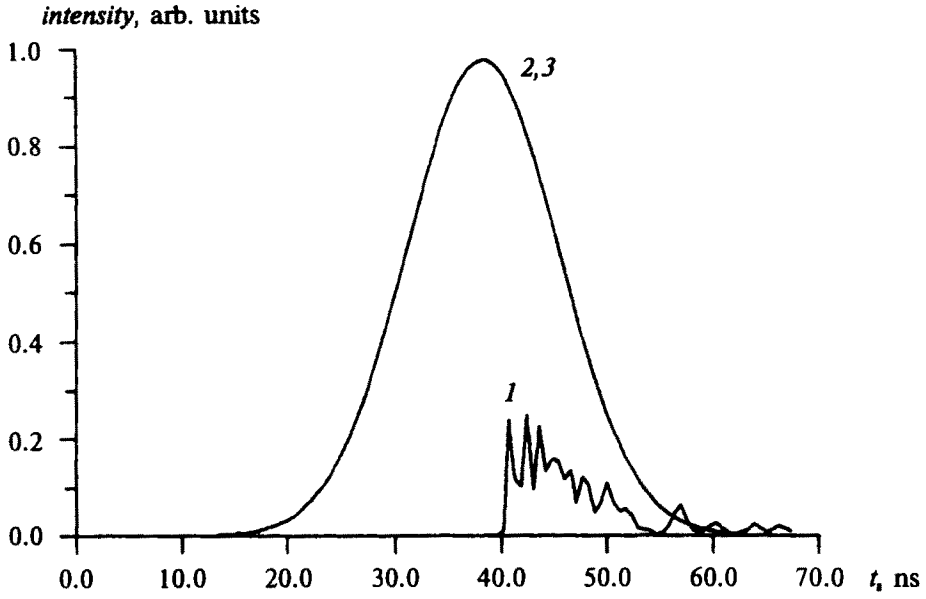


Fig. 7. Shapes of the laser pulses used at the first, second, and third stages of the process at the point $y = 2$ m.

TABLE 3. Results of the mathematical simulation of selective ionization of the ^{168}Yb isotope with quasi-monochromatic laser radiation: n_0 is the natural concentration of ytterbium isotopes; K is the ionization coefficient; n_+ is the relative ion concentration upon the passage of laser radiation

n_0, %	0.14	3.03	14.31	21.82	16.13	31.84	12.73
K, %	46.0	0.00408	0.37724	0.00018	0.00182	0.00006	0.00002
n_+, %	54.182	0.104	45.417	0.034	0.246	0.014	0.0002

Moreover, we performed a complex of calculations in order to determine the coefficient of isotope mixture separation. For this purpose, the center of the laser radiation line was fixed at each stage of the process at the center of spectral lines of the ^{168}Yb isotope, and the ionization coefficients were calculated for all the isotopes. Then we determined the relative concentration of ions and calculated the coefficient of isotope mixture separation. The results of numerical experiment for ^{168}Yb isotope are presented in Table 3. The ionization coefficient was found to have a value of 843.5. It is clear from the analysis of the isotopic structure of transitions at the first and second stages of the process that the selectivity of the process is determined mainly by the second stage.

Thus, in view of the fact that the linewidths of transitions are rather large, the coefficient of isotope mixture separation was found to be high (843). This result considerably exceeds the potentials of conventional methods. Although the ionization is not highly efficient, it should be noted that an output of 10 mg/h is suitable for practical applications. This value can be increased by increasing the efficiency of photoionizing radiation (for comparison, the output of an electromagnetic separator equals 0.26 mg/h).

References

1. J. I. Davis and E. B. Rockower, *IEEE J. Quantum Electron.*, **18**, 233 (1982).
2. W. Martin, R. Zalubas, and L. Hagan, *Atomic Energy Levels of the Rare Earth Elements*, US Government Printing Office, Washington (1978).
3. J. H. Broadhurst, M. E. Cage, D. L. Clark, G., W. Greenlees, J. A. R. Griffith, and G. R. Isaak, *J. Phys. B*, **7**, 513 (1974).
4. M. L. Burshtein, Ya. F. Verolainen, V. A. Comarovskii, et al., *Opt. Spektrosk.*, **37**, 617 (1974).
5. *Elementary Processes in Laser Atom Excitation*, *Proceedings of the Institute of General Physics* [in Russian], Nauka, Moscow (1990), Vol. 24.